```
import pandas as pd
import requests
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.model_selection import train_test_split
import pickle
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Representasi koordinat Jakarta
URL = "https://archive-api.open-meteo.com/v1/archive"
params = {
    "latitude": -6.200000,
    "longitude": 106.816666,
    "start_date": "2021-01-01",
    "end_date": "2026-01-01",
    "daily": ["temperature_2m_max", "temperature_2m_min", "precipitation_sum", "wind_speed_10m_max", "relative_humidity_2m_max"]
    "timezone": "Asia/Jakarta"
}

response = requests.get(URL, params=params)
data = response.json()

# Mengubah ke DataFrame Pandas
df = pd.DataFrame(data["daily"])
print(df.head())
```

```
         time  temperature_2m_max  temperature_2m_min  precipitation_sum  \
0  2021-01-01                29.6                23.5               11.2
1  2021-01-02                29.6                23.8                3.6
2  2021-01-03                30.6                24.1                6.0
3  2021-01-04                29.6                23.5                1.7
4  2021-01-05                28.4                23.5               10.1

   wind_speed_10m_max  relative_humidity_2m_max
0                14.6                        97
1                10.5                        93
2                14.8                        95
3                13.0                        98
4                 9.0                        97
```

```
df.head()
```

	time	temperature_2m_max	temperature_2m_min	precipitation_sum	wind_speed_10m_max	relative_humidity_2m_max
0	2021-01-01	29.6	23.5	11.2	14.6	97
1	2021-01-02	29.6	23.8	3.6	10.5	93
2	2021-01-03	30.6	24.1	6.0	14.8	95
3	2021-01-04	29.6	23.5	1.7	13.0	98
4	2021-01-05	28.4	23.5	10.1	9.0	97

Data Assertions

```
# Informasi dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1827 entries, 0 to 1826
Data columns (total 6 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   time                      1827 non-null   object
 1   temperature_2m_max        1827 non-null   float64
 2   temperature_2m_min        1827 non-null   float64
 3   precipitation_sum         1827 non-null   float64
 4   wind_speed_10m_max        1827 non-null   float64
 5   relative_humidity_2m_max  1827 non-null   int64
dtypes: float64(4), int64(1), object(1)
memory usage: 85.8+ KB
```

```
# cek missing values
df.isnull().sum().sort_values(ascending=False)
```

	0
time	0
temperature_2m_max	0
temperature_2m_min	0
precipitation_sum	0
wind_speed_10m_max	0
relative_humidity_2m_max	0

dtype: int64

```
# Baris terduplikat
df.duplicated().sum()
```

```
np.int64(0)
```

```
# memeriksa apakah data kelembapan sesuai pada rentang 0-100
df[(df['relative_humidity_2m_max'] >= 0) | (df['relative_humidity_2m_max'] <= 100)]
```

	time	temperature_2m_max	temperature_2m_min	precipitation_sum	wind_speed_10m_max	relative_humidity_2m_max
0	2021-01-01	29.6	23.5	11.2	14.6	97
1	2021-01-02	29.6	23.8	3.6	10.5	93
2	2021-01-03	30.6	24.1	6.0	14.8	95
3	2021-01-04	29.6	23.5	1.7	13.0	98
4	2021-01-05	28.4	23.5	10.1	9.0	97
...	...	...	...	...	...	...
1822	2025-12-28	32.5	25.9	11.2	14.3	85
1823	2025-12-29	32.9	25.1	1.2	11.9	87
1824	2025-12-30	31.9	24.5	35.4	13.0	97
1825	2025-12-31	28.5	23.6	36.0	12.9	97
1826	2026-01-01	30.2	23.6	9.5	12.0	97

1827 rows × 6 columns

Data Preprocessing

```
df['time'] = pd.to_datetime(df['time'])
df.set_index('time', inplace=True)
```

```
df
```

time	temperature_2m_max	temperature_2m_min	precipitation_sum	wind_speed_10m_max	relative_humidity_2m_max
2021-01-01	29.6	23.5	11.2	14.6	97
2021-01-02	29.6	23.8	3.6	10.5	93
2021-01-03	30.6	24.1	6.0	14.8	95
2021-01-04	29.6	23.5	1.7	13.0	98
2021-01-05	28.4	23.5	10.1	9.0	97
...	...	...	...	...	...
2025-12-28	32.5	25.9	11.2	14.3	85
2025-12-29	32.9	25.1	1.2	11.9	87
2025-12-30	31.9	24.5	35.4	13.0	97
2025-12-31	28.5	23.6	36.0	12.9	97
2026-01-01	30.2	23.6	9.5	12.0	97

1827 rows × 5 columns

```
# membuat feature curah hujan besok
df['precipitation_sum_besok'] = df['precipitation_sum'].shift(-1)
df.head()
```

time	temperature_2m_max	temperature_2m_min	precipitation_sum	wind_speed_10m_max	relative_humidity_2m_max	precipitation_su
2021-01-01	29.6	23.5	11.2	14.6	97	
2021-01-02	29.6	23.8	3.6	10.5	93	
2021-01-03	30.6	24.1	6.0	14.8	95	

```
df.tail()
```

time	temperature_2m_max	temperature_2m_min	precipitation_sum	wind_speed_10m_max	relative_humidity_2m_max	precipitation_su
2025-12-28	32.5	25.9	11.2	14.3	85	
2025-12-29	32.9	25.1	1.2	11.9	87	
2025-12-30	31.9	24.5	35.4	13.0	97	

```
# membuat kolom hujan besok
df['hujan_besok'] = (df['precipitation_sum_besok'] > 0.5).astype(int)
```

```
df
```

	temperature_2m_max	temperature_2m_min	precipitation_sum	wind_speed_10m_max	relative_humidity_2m_max	precipitation_sur
time						
2021-01-01	29.6	23.5	11.2	14.6	97	
2021-01-02	29.6	23.8	3.6	10.5	93	
2021-01-03	30.6	24.1	6.0	14.8	95	
2021-01-04	29.6	23.5	1.7	13.0	98	
2021-01-05	28.4	23.5	10.1	9.0	97	
...	...	...	...	...	...	
2025-12-28	32.5	25.9	11.2	14.3	85	
2025-12-29	32.9	25.1	1.2	11.9	87	

```
df.dropna(inplace=True)
```

```
df.drop(columns=['precipitation_sum_besok'], inplace=True)
```

```
df
```

	temperature_2m_max	temperature_2m_min	precipitation_sum	wind_speed_10m_max	relative_humidity_2m_max	precipitation_sur
time						
2021-01-01	29.6	23.5	11.2	14.6	97	
2021-01-02	29.6	23.8	3.6	10.5	93	
2021-01-03	30.6	24.1	6.0	14.8	95	
2021-01-04	29.6	23.5	1.7	13.0	98	
2021-01-05	28.4	23.5	10.1	9.0	97	
...	...	...	...	...	...	
2025-12-27	32.0	25.7	1.0	18.5	87	
2025-12-28	32.5	25.9	11.2	14.3	85	

```
df.isnull().sum().sort_values(ascending=False)
```

	0
temperature_2m_max	0
temperature_2m_min	0
precipitation_sum	0
wind_speed_10m_max	0
relative_humidity_2m_max	0
precipitation_sum_besok	0
hujan_besok	0

dtype: int64

EDA

```
df['hujan_besok'].value_counts(normalize=True)
```

hujan_besok	proportion
1	0.836802
0	0.163198

dtype: float64

```
df['hujan_besok'].value_counts()
```

hujan_besok	count
1	1528
0	298

dtype: int64

```
# korelasi antar feature dengan target
df.corr()['hujan_besok'].sort_values(ascending=False)
```

	hujan_besok
hujan_besok	1.000000
precipitation_sum_besok	0.360638
relative_humidity_2m_max	0.269290
precipitation_sum	0.239391
temperature_2m_min	0.138741
wind_speed_10m_max	-0.153497
temperature_2m_max	-0.279255

dtype: float64

Training Preparation

```
X = df.drop(columns=['hujan_besok'])
y = df['hujan_besok']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y, test_size=0.2, random_state=42)
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Model Training

```
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

Model Evaluation

```
# classfication report
y_pred = model.predict(X_test_scaled)
print(classification_report(y_test, y_pred))
```

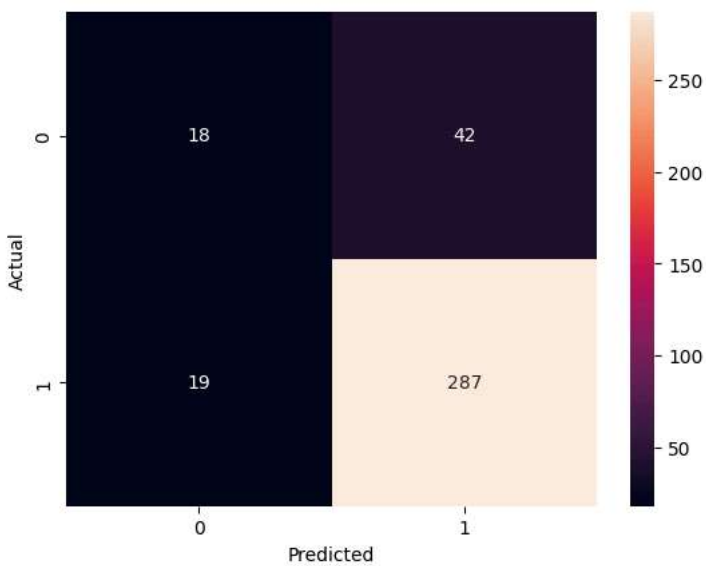
```
              precision    recall  f1-score   support

           0       0.49      0.30      0.37        60
           1       0.87      0.94      0.90       306

    accuracy                           0.83       366
   macro avg       0.68      0.62      0.64       366
weighted avg       0.81      0.83      0.82       366
```

```
# confusion matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel('Predicted')
plt.ylabel('Actual')
```

```
Text(50.722222222222214, 0.5, 'Actual')
```




```
# export model
with open('trained_rf.pkl', 'wb') as f:
    pickle.dump(model, f)
```

```
# export scaler
with open('scaler.pkl', 'wb') as f:
    pickle.dump(scaler, f)
```